(KTXFI2EBNF) Physics II. Exercises

December 18, 2025

1. By how much does the emitted current change if the tungsten cathode temperature is increased from $T = 2200\,\mathrm{K}$ to $2350\,\mathrm{K}$? The work function for tungsten is $W_{\mathrm{ki}} = 4.5\,\mathrm{eV}$. The Richardson coefficient is $6.02 \cdot 10^5\,\mathrm{A/m^2K^2}$.

2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at $T = 2200\,\mathrm{K}$? The electron charge is $q = -1.6 \cdot 10^{-19}\,\mathrm{As}$.

3. What is the free-electron drift velocity in a copper conductor of radius $R = 1.5\,\mathrm{mm}$ if the current is $I = 12\,\mathrm{A}$? The electron density in copper is $n_{\mathrm{e}} = 3.3 \cdot 10^{28}\,\mathrm{m^{-3}}$.

4. We examine an n-type semiconductor under the conditions shown in the figure: $B_y = 0.4\,\mathrm{T}$, $U_{\mathrm{H}} = 10\,\mathrm{mV}$, $I = 30\,\mathrm{mA}$. The magnetic induction is directed along the y–axis.

   (a) What is the electron concentration?

   (b) What is the drift velocity of the electrons?

   (c) What is the Hall constant?

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